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Base member

Abstract

A base member includes a via having a first hole and a second hole. When viewed from a direction in which a base surface extends, the first hole and the second hole have shapes becoming wider as getting closer to a first surface and a second surface of the base surface from a portion at which a lower end of the first hole and an upper end of the second hole are in contact with each other. When the first hole and the second hole are viewed from the direction in which the base surface extends, a smaller angle of angles formed by a generatrix of the first hole and a generatrix of the second hole is smaller in a thin region of the base member than in a thick region of the base member.

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Background/Summary

BACKGROUND
Field
(1) The present disclosure relates to a base member formed using a molded interconnect device (MID) technique.
Description of the Related Art
(2) Recent electronic devices, such as cameras and smartphones, and transport devices, such as automobiles, have incorporated therein sensor modules that are modularized and equipped with various sensors.
(3) As such sensor modules, sensor modules formed using the MID technique have been known.
(4) The MID technique is a technique of irradiating a predetermined portion of a base member with laser and coating the irradiated portion with metal plating. The portion coated with metal plating functions as a conductive pattern.
(5) For example, Japanese Patent Application Laid-Open No. 2013-44645 discusses an MID package including three gyroscopic elements provided on the respective three slopes of a main body having a substantially three-sided pyramid shape in which a first pyramidal surface, a second pyramidal surface, and a third pyramidal surface are orthogonal to each other.
(6) In recent years, there has been an increasing demand for the downsizing of electronic devices.

Accordingly, the realization of a base member like a compact sensor module that can be easily accommodated into a downsized electronic device has been desired.
(7) With the conventional technique discussed in Japanese Patent Application Laid-Open No. 2013-44645, large dimensions of the MID package in longitudinal, transverse, and thickness directions have hindered the downsizing of electronic devices.
(8) In addition, the three-sided pyramid shape has been undesirable from the aspect of accommodating the MID package into an electronic device because a dead space is easily generated inside the electronic device.
SUMMARY
(9) The present disclosure is directed to providing a compact base member having a shape suitable for being accommodated in an electronic device.
(10) According to an aspect of the present disclosure, a base member is a molded component in which pattern wiring is directly formed. The base member includes a base surface including a first surface and a second surface on an opposite side of the first surface, and a via for electrically connecting the first surface and the second surface. In the via, a shape at a lower end of a first hole having a circular frustum shape and a shape at an upper end of a second hole having a circular frustum shape are same as each other. When viewed from a direction in which the base surface extends, the via has a structure in which a pattern is formed from the first surface to the second surface of the base surface on an internal wall of a hole having a shape in which the first hole and the second hole are stacked in such a manner that the lower end of the first hole and the upper end of the second hole are in contact with each other. When viewed from the direction in which the base surface extends, the first hole and the second hole have the circular frustum shapes becoming wider as getting closer to the first surface and the second surface of the base surface from a portion at which the lower end of the first hole and the upper end of the second hole are in contact with each other. When the first hole and the second hole are viewed from the direction in which the base surface extends, a smaller angle of angles formed by a generatrix of the first hole and a generatrix of the second hole is smaller in a thin region of the base member than in a thick region of the base member.
(11) Further features of the present disclosure will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a block diagram illustrating a configuration of a camera according to a first exemplary embodiment of the present disclosure.
(2) FIGS. **2**A and **2**B are external perspective views of the camera according to the first exemplary embodiment.
(3) FIG. **3** is an exploded perspective view of the camera according to the first exemplary embodiment.
(4) FIGS. **4**A and **4**B are perspective views illustrating an internal configuration of a top cover unit according to the first exemplary embodiment.
(5) FIG. **5**A is an exploded perspective view of a shake detection unit according to the first exemplary embodiment. FIGS. **5**B and **5**C are side views of the shake detection unit according to the first exemplary embodiment.
(6) FIGS. **6**A and **6**B are external perspective views of a sensor module according to the first exemplary embodiment. FIG. **6**C is a cross-sectional view of a main part of the sensor module according to the first exemplary embodiment. FIG. **6**D is a plan view of a land shape according to the first exemplary embodiment. FIG. **6**E is a cross-sectional perspective view of a via portion

according to the first exemplary embodiment. FIG. **6**F is a cross-sectional enlarged view of the via portion according to the first exemplary embodiment.
(7) FIGS. **7**A and **7**B are side views of the sensor module according to the first exemplary embodiment.
(8) FIGS. **8**A and **8**B are a side view and a perspective view of a sensor module according to a second exemplary embodiment, respectively. FIG. **8**C is a cross-sectional view of a via according to the second exemplary embodiment.
(9) FIG. **9**A is a cross-sectional enlarged view of a via according to a third exemplary embodiment. FIGS. **9**B and **9**C are cross-sectional perspective views of the via according to the third exemplary embodiment. FIG. **9**D is a schematic view of pattern wiring according to the third exemplary embodiment.
(10) FIGS. **10**A and **10**B are a cross-sectional perspective view and a cross-sectional view of a via according to a fourth exemplary embodiment, respectively.
(11) FIG. **11** is a plan view of a land shape according to a fifth exemplary embodiment.
(12) FIG. **12** is a plan view of a land shape according to a sixth exemplary embodiment.
(13) FIG. **13** is a plan view of a land shape according to a seventh exemplary embodiment.
(14) FIGS. **14**A and **14**B are a side view and a perspective view of a shake detection unit according to an eighth exemplary embodiment, respectively.
DESCRIPTION OF THE EMBODIMENTS
(15) Hereinafter, exemplary embodiments of the present disclosure will be described in detail with reference to the attached drawings.
(16) A molded interconnect device (MID) refers to an injection molded component where wiring and electrodes are formed.
(17) A first exemplary embodiment of the present disclosure will be described with reference to FIGS. **1** to **7**B.
(18) <Block Diagram Illustrating Configuration Example of Digital Camera>
(19) FIG. **1** is a block diagram illustrating a configuration example of a digital camera (hereinafter referred to as a camera) **100** according to the present exemplary embodiment. The camera **100** is an interchangeable lens camera to which a lens unit **200** is detachably attached.
(20) FIGS. **2**A and **2**B are external perspective views of the camera **100**.
(21) The lens unit **200** illustrated in FIG. **1** is fixed to the camera **100** by a lens mount **201** included in the lens unit **200** and a lens mount **101** included in the camera **100**.
(22) The lens unit **200** and the camera **100** are configured to communicate with each other via a connector **202** included in the lens unit **200** and a connector **102** included in the camera **100**.
(23) More specifically, a system control unit **307** and a lens drive control unit **203** communicate with each other.
(24) Based on a signal from the system control unit **307**, the lens drive control unit **203** controls a lens drive unit **204** to drive a diaphragm **211** and a lens **210**.
(25) The lens **210** is configured to form an optical image of a subject onto an image sensor **302**.
(26) A shutter **301** is arranged between the image sensor **302** and the lens **210**, and blocks light from the lens **210** to the image sensor **302** in a non-image-capturing state.
(27) At the time of image capturing, the shutter **301** is opened under control of the system control unit **307**, so that the lens **210** can form the optical image onto the image sensor **302**.
(28) The image sensor **302** is for example a charge-coupled device (CCD) image sensor or a complementary metal-oxide semiconductor (CMOS) image sensor that converts the optical image into an electrical signal.
(29) An analog-to-digital (A/D) converter **304** converts an analog signal into a digital signal. The A/D converter **304** is used to convert the analog signal output from the image sensor **302**, into a digital signal.
(30) An image processing unit **305** performs resizing processing, such as predetermined pixel

interpolation or reduction, and color conversion processing on data from the A/D converter **304** or data from a memory control unit **306**.
(31) The image processing unit **305** also performs predetermined calculation processing using data of a captured image. The system control unit **307** performs exposure control and ranging control based on a calculation result obtained by the calculation processing.
(32) Through-the-lens (TTL) system autofocus (AF) processing, autoexposure (AE) processing, and electronic flash pre-emission (EF) processing are thereby performed.
(33) Furthermore, the image processing unit **305** performs predetermined calculation processing using data of a captured image and performs TTL system automatic white balance (AWB) processing based on a calculation result obtained by the calculation processing.
(34) Output data from the A/D converter **304** is directly written into a memory **308** via the image processing unit **305** and the memory control unit **306**, or via the memory control unit **306**.
(35) The memory **308** stores image data for displaying image data subjected to the A/D conversion by the A/D converter **304**, on a display unit **105** or **106**.
(36) The memory **308** also serves as a memory (a video memory) for image display.
(37) A digital-to-analog (D/A) converter **309** converts the data for image display that is stored in the memory **308**, into an analog signal, and supplies the analog signal to the display unit **105** or **106**.
(38) In this manner, the image data to be displayed, which has been written into the memory **308**, is displayed by the display unit **105** or **106** via the D/A converter **309**.
(39) The display units **105** and **106** perform display based on the analog signal from the D/A converter **309**, on a display device such as a liquid crystal display (LCD).
(40) The digital signals subjected to the A/D conversion once by the A/D converter **304** and stored in the memory **308** are subjected to the D/A conversion by the D/A converter **309**.
(41) The resulting analog signals are then sequentially transferred to the display unit **105** or **106** and displayed thereon, so that through-the-lens image display (live view display) can be performed.
(42) A nonvolatile memory **310** serves as an electrically erasable and recordable recording medium. For example, an electrically erasable programmable read-only memory (EEPROM) is used as the nonvolatile memory **310**.
(43) The nonvolatile memory **310** stores constants for operating the system control unit **307**, programs, and the like.
(44) The system control unit **307** includes at least one processor, and controls the entire camera **100** and the lens unit **200**.
(45) A random access memory (RAM) is used as a system memory **311**. Constants and variables for operating the system control unit **307**, and the programs read out from the nonvolatile memory **310** are loaded into the system memory **311**.
(46) The system control unit **307** also performs display control by controlling the memory **308**, the D/A converter **309**, and the display unit **105** or **106**.
(47) A system timer **312** measures times used in various types of control or a time of a built-in clock.
(48) A first shutter switch **104***a* is turned on in the middle of an operation of a shutter button **104** provided on the camera **100**, i.e., when the shutter button **104** is pressed halfway (to issue an image capturing preparation instruction), and generates a first shutter switch signal SW**1**.
(49) Based on the first shutter switch signal SW**1**, operations such as AF processing, AE processing, AWB processing, and EF processing are started.
(50) A second shutter switch **104***b* is turned on upon completion of the operation of the shutter button **104**, i.e., when the shutter button **104** is pressed completely (to issue an image capturing instruction), and generates a second shutter switch signal SW**2**.
(51) Based on the second shutter switch signal SW**2**, the system control unit **307** controls the shutter **301** to drive a shutter blade **301***a*.

(52) The system control unit **307** then starts operations of a series of image capturing processes from the readout of a signal from the image sensor **302** to the writing of image data onto a recording medium **330**.
(53) The shutter blade **301***a* operates at a high speed inside the shutter **301** in a direction vertical to an optical axis of the lens **210**, and instantaneously stops the operation by hitting a stopper member (not illustrated) provided inside the shutter **301**.
(54) When various function icons displayed on the display unit **105** or **106** are selectively operated, functions are appropriately allocated to operation members of an operation unit **108** depending on the scene, so that the operation members serve as various function buttons.
(55) Examples of the function buttons include an end button, a return button, an image feed button, a jump button, a narrowing down button, and an attribute change button.
(56) For example, if a menu button is pressed, a menu screen for enabling a user to make various settings is displayed on the display unit **105** or **106**.
(57) A power switch **103** is used to power on or off the camera **100**.
(58) A power source control unit **313** includes a battery detection circuit, a direct current (DC)-DC converter, and a switch circuit for switching a block to be supplied with power. The power source control unit **313** detects whether a battery is attached, a type of the battery, and a remaining capacity of the battery.
(59) In addition, the power source control unit **313** controls the DC-DC converter based on a result of the detection and an instruction from the system control unit **307**, and supplies an appropriate voltage for an appropriate time period to the components including the recording medium **330**.
(60) A power source unit **314** includes a primary battery such as an alkaline battery or a lithium battery, a secondary battery such as a nickel-cadmium (NiCd) battery, a nickel-metal hydride (NiMH) battery, or a lithium (Li) battery, and an alternating current (AC) adapter.
(61) A recording medium interface (I/F) **315** is an interface with the recording medium **330** such as a memory card or a hard disc.
(62) The recording medium **330** is for example a memory card for recording a captured image, and includes a semiconductor memory, an optical disc, or a magnetic disc.
(63) A communication unit **316** is connected wirelessly or via a wired cable, and transmits or receives a video signal, an audio signal, and the like.
(64) The communication unit **316** is also connectable with a wireless local area network (LAN) and the Internet.
(65) The communication unit **316** is capable of transmitting an image (including a through-the-lens image) captured by the image sensor **302** and an image recorded on the recording medium **330**. The communication unit **316** is also capable of receiving image data and other various types of information from an external device.
(66) <Shake Detection Unit/Gyro Sensor>
(67) A shake detection unit **320** is for example a gyro sensor, and detects a shake amount of the camera **100**.
(68) The shake detection unit **320** detects the shake and the shake amount of the camera **100** in three axis directions, i.e., a pitch direction, a yaw direction, and a roll direction.
(69) In the camera **100** illustrated in FIG. **1**, an image sensor drive unit **303** performs optical shake correction by controlling a movement of the image sensor **302** based on the shake amount detected by the shake detection unit **320**.
(70) Under control of the system control unit **307**, the image processing unit **305** electronically performs shake correction of an image based on the shake amount detected by the shake detection unit **320**.
(71) <Exploded Perspective View of Camera>
(72) FIG. **3** is an exploded perspective view of the camera **100** excluding front, rear, and lower cover members.

(73) A ground plate **120** is a structure that provides strength of the camera **100**. The shutter **301**, the image sensor **302**, the image sensor drive unit **303**, and the system control unit **307** are fastened to the ground plate **120** with screws (not illustrated).
(74) A top cover unit **110** is fastened to the ground plate **120** with screws (not illustrated).
(75) <External Perspective View of Top Cover Unit>
(76) FIG. **4**A is an external perspective view of the top cover unit **110**. FIG. **4**B is an exploded perspective view of the top cover unit **110**.
(77) The top cover unit **110** includes a top cover member **111** covering a top surface of the camera **100**, the power switch **103**, and the shutter button **104**.
(78) As illustrated in FIG. **4**B, the shake detection unit **320** is fastened to a center portion of the top cover member **111** with screws **401**.
(79) <Exploded Perspective View of Shake Detection Unit>
(80) FIG. **5**A is an exploded perspective view of the shake detection unit **320**. FIGS. **5**B and **5**C are external perspective views of the shake detection unit **320**.
(81) The shake detection unit **320** includes plates **505** and **506**, two buffer members **504**, a sensor module **508**, and a flexible substrate **502**F.
(82) The sensor module **508** includes gyro sensors **501**P, **501**R, and **501**Y for measuring an angular velocity, and an acceleration sensor **502**A for detecting acceleration.
(83) The sensor module **508** further includes passive elements **502**R (refer to FIG. **6**B), such as a resistor and a capacitor, for driving the three gyro sensors **501**P, **501**R, and **501**Y and the acceleration sensor **502**A.
(84) The flexible substrate **502**F includes terminal portions **502**D and **502**E.
(85) The terminal portion **502**D is connected to a connector **502**C (refer to FIG. **6**B) included in the sensor module **508**.
(86) The terminal portion **502**E is electrically connected with a connector (not illustrated) forming a part of a path to the system control unit **307**.
(87) With this configuration, the sensor module **508** and the system control unit **307** are capable of appropriately performing various types of communication via the flexible substrate **502**F.
(88) Through this communication, the system control unit **307** is capable of acquiring detection values detected by the gyro sensors **501**P, **501**R, and **501**Y and the acceleration sensor **502**A.
(89) The two buffer members **504** are formed of an elastic material, such as a sponge, that absorbs vibration.
(90) The shake detection unit **320** includes the two buffer members **504**.
(91) The two buffer members **504** are arranged on different surfaces facing the sensor module **508** to sandwich the sensor module **508**.
(92) In the present exemplary embodiment, the two buffer members **504** are arranged on the surfaces substantially parallel to a primary surface **503**Y that is a surface on which the gyro sensor **501**Y is arranged.
(93) The plates **505** and **506** include the surfaces substantially parallel to the primary surface **503**Y. The plates **505** and **506** sandwich the sensor module **508** and the two buffer members **504**.
(94) The plate **505** and the sensor module **508** compress one of the two buffer members **504**, and the plate **506** and the sensor module **508** compress the other buffer member **504**.
(95) The plates **505** and **506** are thus configured not to make contact with the sensor module **508**.
(96) The shake detection unit **320** is completed by fastening of the plates **505** and **506** using screws **507**.
(97) <External Perspective View of Sensor Module>
(98) FIGS. **6**A and **6**B are external perspective views of the sensor module **508**. FIG. **6**C is a cross-sectional view of a main part of the sensor module **508**.
(99) A base member **503** illustrated in FIG. **6**A holds the gyro sensors **501**P, **501**R, and **501**Y at predetermined positions, and is molded using a material such as plastic or liquid crystal polymer

(LCP).
(100) The base member **503** includes a first side wall **503**P substantially orthogonal to a rotational axis of the pitch direction, a second side wall **503**R substantially orthogonal to a rotational axis of the roll direction, and the primary surface **503**Y substantially orthogonal to a rotational axis of the yaw direction.
(101) In other words, the first side wall **503**P is orthogonal to the primary surface **503**Y, and the second side wall **503**R is orthogonal to each of the primary surface **503**Y and the first side wall **503**P.
(102) The gyro sensor **501**P is mounted on the first side wall **503**P, the gyro sensor **501**R is mounted on the second side wall **503**R, and the gyro sensor **501**Y is mounted on the primary surface **503**Y.
(103) The gyro sensors **501**P, **501**R, and **501**Y are directly soldered to land portions of pattern wiring **500** on the base member **503**.
(104) The sensor module **508** has the above-described configuration.
(105) The gyro sensor **501**P thus detects the shake and the shake amount of the camera **100** in the pitch direction, the gyro sensor **501**R detects the shake and the shake amount of the camera **100** in the roll direction, and the gyro sensor **501**Y detects the shake and the shake amount of the camera **100** in the yaw direction.
(106) The connector **502**C is mounted on a surface of the primary surface **503**Y that is opposite to a surface of the primary surface **503**Y on which the gyro sensor **501**Y is mounted.
(107) Hereinafter, the surface of the primary surface **503**Y on which the gyro sensor **501**Y is mounted will be referred to as a first surface **509**, and the surface of the primary surface **503**Y on which the connector **502**C is mounted will be referred to as a second surface **510**.
(108) The acceleration sensor **502**A is also mounted on the second side wall **503**R.
(109) As illustrated in FIGS. **6**A and **6**B, the pattern wiring **500** is directly formed on the base member **503**.
(110) The pattern wiring **500** is formed using a molded interconnect device (MID) technique.
(111) The MID technique is a technique of forming a pattern by irradiating a predetermined portion of a base member with laser, and coating the irradiated portion with metal plating.
(112) The gyro sensors **501**P, **501**R, and **501**Y and the acceleration sensor **502**A for shake detection, and the passive elements **502**R for driving these sensors will be described next.
(113) The gyro sensors **501**P, **501**R, and **501**Y, the acceleration sensor **502**A, and the passive elements **502**R are directly mounted by soldering on the mounting land portions provided by the pattern wiring **500** on the base member **503**.
(114) The pattern wiring **500** enables appropriate electrical connection between the gyro sensors **501**P, **501**R, and **501**Y, the acceleration sensor **502**A, the passive elements **502**R, and the connector **502**C.
(115) FIG. **6**C is a cross-sectional view of the main part of the sensor module **508**.
(116) Assuming that a thickness of the primary surface **503**Y is Z, and the shorter one of lengths of the first side wall **503**P and the second side wall **503**R in a direction orthogonal to the primary surface **503**Y is W, the length W is larger than the thickness Z.
(117) Assume that the higher one of a height of the first side wall **503**P from the first surface **509** and a height of the second side wall **503**R from the first surface **509** is A, and the higher one of a height of the first side wall **503**P from the second surface **510** and a height of the second side wall **503**R from the second surface **510** is B.
(118) In addition, assume that a height from the first surface **509** to a top surface of the gyro sensor **501**Y is C, and a height from the second surface **510** to a top surface of the connector **502**C is D.
(119) In this case, the height A is equal to or larger than the height C, and the height B is equal to or larger than the height D.
(120) In the present exemplary embodiment, the above-described configuration is employed.

(121) Thus, the gyro sensor **501**Y and the connector **502**C do not protrude from the higher one of the first side wall **503**P and the second side wall **503**R in a direction vertical to the primary surface **503**Y.
(122) Therefore, a height of the sensor module **508** in the direction orthogonal to the primary surface **503**Y can be reduced.
(123) FIG. **6**D is an enlarged view of lands **522** for mounting the gyro sensor **501**P that are formed on the first side wall **503**P.
(124) The gyro sensor **501**P is mounted onto the first side wall **503**P by reflow mounting using solder.
(125) In FIG. **6**D, an arrow G indicates a gravity direction (a direction G) in melting a solder paste in the reflow mounting of the gyro sensor **501**P on the first side wall **503**P.
(126) In addition, an arrow H indicates a direction (a direction H) that is orthogonal to the gravity direction and parallel to the first side wall **503**P.
(127) In normal reflow mounting on a flat-shaped substrate, gravity constantly acts on a component to be mounted, vertically with respect to a surface on which the component is to be mounted.
(128) However, in the shake detection unit **320** according to the present exemplary embodiment, the gyro sensors **501**P, **501**R, and **501**Y are mounted on the respective three surfaces (the first side wall **503**P, the second side wall **503**R, and the primary surface **503**Y) that are orthogonal to each other.
(129) Accordingly, in the reflow mounting, the gravity acts on at least one of the gyro sensors **501**P, **501**R, and **501**Y in a direction parallel to the mounting surface.
(130) In the present exemplary embodiment, the gyro sensor **501**P is mounted onto the first side wall **503**P in a state where the gravity acts in the direction indicated by the arrow G.
(131) Thus, when the solder melts, the mounting position may shift from a desired mounting position in the direction indicated by the arrow G, under the influence of the gravity in the direction indicated by the arrow G.
(132) In view of the foregoing, in land arrays A and B each including five lands **522**, the amount of solder to be applied to the land array B on the lower side in FIG. **6**D is made smaller than the amount of solder to be applied to the land array A on the upper side.
(133) As a result, surface tension of the solder applied to the land array A is larger than surface tension of the solder applied to the land array B.
(134) With this configuration, it is possible to prevent mounting misalignment caused by the gravity in the reflow mounting, by changing the direction of the force acting on the gyro sensor **501**P, to a reverse direction of the direction indicated by the arrow G, using the surface tension of the solder, and cancelling out the gravity acting on the gyro sensor **501**P in the direction indicated by the arrow G.
(135) As illustrated in FIGS. **6**A and **6**B, the base member **503** is provided with a plurality of vias **520** for electrically connecting one surface and the other surface of the primary surface **503**Y.
(136) FIG. **6**E is an enlarged cross-sectional perspective view of the base member **503** in a portion where the vias **520** are arranged. FIG. **6**F is an enlarged cross-sectional view of the base member **503** in the portion where the vias **520** are arranged.
(137) For improving productivity, a structure of a mold for molding the base member **503** may be such that the base member **503** is demolded in the direction vertical to the primary surface **503**Y because the primary surface **503**Y is the largest planar surface.
(138) For this reason, in the present exemplary embodiment, arranging all the vias **520** on the primary surface **503**Y enables a direction of a hole **512** of each of the vias **520** to correspond to the demolding direction, thereby simplifying the mold structure.
(139) In the present exemplary embodiment, as illustrated in FIGS. **6**E and **6**F, the hole **512** of each of the vias **520** is formed into a shape in which two frustums are stacked symmetrically about a center in a thickness direction of the primary surface **503**Y.

(140) Each of the frustums has a shape in which the diameter is smallest at the center in the thickness direction of the primary surface **503**Y, and becomes larger in a direction away from the center in the thickness direction.

(141) Assuming that a diameter of a circle at a via vertex **515** is a via diameter **514**, the hole **512** is opened up to the surfaces of the base member **503** with the via diameter **514** increasing at a fixed via angle **513** starting from the via vertex **515**, and penetrates through the base member **503**.

(142) On an internal wall of the hole **512**, a pattern is directly wired from one surface to the other surface of the primary surface **503**Y, using the MID technique.

(143) Accordingly, patterns wired on one surface and the other surface of the base member **503** are electrically connected via the vias **520**.

(144) The hole **512** according to the present exemplary embodiment has the above-described shape instead of a simple cylindrical shape penetrating through the primary surface **503**Y, in order to facilitate the laser irradiation of the internal wall of the hole **512** for pattern formation.

(145) <Side View of Sensor Module>

(146) FIGS. **7**A and **7**B are side views of the sensor module **508**. FIG. **7**A illustrates the first side wall **503**P, and FIG. **7**B illustrates the second side wall **503**R.

(147) The gyro sensors **501**Y, **501**R, and **501**P according to the present exemplary embodiment are substantially rectangles when viewed from directions vertical to the respective surfaces (the primary surface **503**Y, the second side wall **503**R, and the first side wall **503**P) on which the gyro sensors **501**Y, **501**R, and **501**P are mounted.

(148) Thus, the gyro sensor **501**R has long sides **501**Ra and short sides **501**Rb, and the gyro sensor **501**P has long sides **501**Pa and short sides **501**Pb.

(149) In the present exemplary embodiment, the gyro sensors **501**R and **501**P are attached to the base member **503** in such a manner that the short sides **501**Rb and the short sides **501**Pb are parallel to the direction orthogonal to the primary surface **503**Y.

(150) The heights of the first side wall **503**P and the second side wall **503**R in the direction orthogonal to the primary surface **503**Y can be accordingly reduced.

(151) Definition of a frustum according to an exemplary embodiment of the present disclosure will be described now.

(152) The frustum is a three-dimensional object obtained by removing, from a cone or pyramid, a similar reduced cone or pyramid having the same vertex as that of the cone or pyramid. In other words, the frustum is a three-dimensional object surrounded by a conical or pyramidal surface and two parallel planar surfaces.

(153) A frustum formed from a circular cone will be referred to as a circular frustum, a frustum formed from a pyramid will be referred to as a pyramid frustum, and a frustum formed from an n-sided pyramid will be referred to as an n-sided frustum. A pyramid such as a three-sided pyramid, a four-sided pyramid, a six-sided pyramid, or an eight-sided pyramid can be applied as the n-sided pyramid.

(154) In a second exemplary embodiment to be described below, a more desirable circular frustum will be used as an example.

(155) The present exemplary embodiment will be described with reference to FIGS. **8**A to **8**C.

(156) FIG. **8**A is a side view of a sensor module **900**.

(157) FIG. **8**B is an external perspective view of the sensor module **900**.

(158) FIG. **8**C is a cross-sectional view of a via arranged in a second portion **903** of a primary surface **901** of a base member **904** of the sensor module **900**.

(159) The sensor module **900** is configured to mount the gyro sensors **501**Y, **501**R, and **501**P on the base member **904**.

(160) A thickness of a first portion **902** of the primary surface **901** of the base member **904** is the same as that of the primary surface **503**Y of the base member **503** according to the first exemplary embodiment.

(161) A thickness of the second portion **903** of the primary surface **901** is thicker than the thickness of the primary surface **503**Y of the base member **503** according to the first exemplary embodiment. With this configuration, the base member **904** according to the present exemplary embodiment has higher strength than that of the base member **503** according to the first exemplary embodiment.
(162) In FIGS. **8**A to **8**C, illustration of the pattern wiring **500** and the passive elements **502**R such as a resistor and a capacitor is omitted for ease of understanding.
(163) The primary surface **901** of the base member **904** includes a plurality of the vias (the first vias) **520** having a first shape and a plurality of second vias **905** having a second shape, which are provided to electrically connect one surface and the other surface of the primary surface **901**.
(164) The first vias **520** having the first shape are formed in the first portion **902**, and the second vias **905** having the second shape are formed in the second portion **903**.
(165) The first shape of the first vias **520** is the same as the shape described in the first exemplary embodiment, and the description thereof will thus be omitted.
(166) The second vias **905** and the first vias **520** have similar shapes in that each hole **908** is formed into a shape in which two frustums are stacked, but differ in that a via angle **906** is larger than the via angle **513** of each of the first vias **520**.
(167) On an internal wall of the hole **908**, a pattern is directly wired from one surface to the other surface of the primary surface **901**, using the MID technique.
(168) Thus, patterns wired on one surface and the other surface of the base member **904** are electrically connected via the second vias **905**.
(169) If the sensor module **900** is mounted on the camera **100** and is operated, a temperature of the sensor module **900** rises during the operation and drops when the operation stops.
(170) The rise and drop of the temperature causes the base member **904** of the sensor module **900** to repeat thermal expansion and thermal contraction.
(171) The expansion and contraction of the base member **904** causes stress to be applied to the pattern formed on the internal wall of the hole **908**.
(172) In particular, large stress is applied to the pattern at a via vertex **909** due to stress concentration, which causes the pattern to break.
(173) As a thickness of the primary surface **901** increases, the stress applied to the pattern at the via vertex **909** increases. On the other hand, as a value of the via angle **906** increases, the stress applied to the pattern at the via vertex **909** decreases.
(174) In the present exemplary embodiment, a case where the first vias **520** having the first shape are arranged in the second portion **903** having a large thickness will be considered.
(175) In this case, large stress is applied to the pattern at the via vertex **515** as compared with the case where the first vias **520** are arranged in the first portion **902**, and this can cause the pattern to break on the internal wall.
(176) For this reason, the second vias **905** having the second shape with which a reduced stress at the via vertex **909** can be expected more than the first vias **520** having the first shape are arranged in the second portion **903**.
(177) If the via angle **906** is changed to an obtuse angle, laser irradiation of the internal wall of each of the holes **908** for pattern formation can be difficult.
(178) Thus, to facilitate the laser irradiation of the internal wall of the hole **908**, a via diameter **907** can be increased.
(179) The present exemplary embodiment will be schematically described next.
(180) The base member **904** is a molded component in which the pattern wiring **500** is directly formed.
(181) The base member **904** includes the second vias **905** for electrically connecting one surface of the primary surface (the base surface) **901** of the base member **904**, and the other surface of the primary surface (the base surface) **901** on an opposite side of the one surface.
(182) In each of the second vias **905**, a shape at a lower end of a first hole having a circular frustum

shape and a shape at an upper end of a second hole having a circular frustum shape are the same as each other.
(183) Each of the second vias **905** has a structure in which a pattern is formed from one surface to the other surface of the primary surface (the base surface) **901** on the internal wall of the hole **908** having a shape in which the first hole and the second hole are stacked in such a manner that the lower end of the first hole and the upper end of the second hole are in contact with each other.
(184) The first hole and the second hole have the circular frustum shapes becoming wider as getting closer to one surface and the other surface of the primary surface (the base surface) **901** from a portion at which the lower end of the first hole and the upper end of the second hole are in contact with each other.
(185) The smaller one (which is twice as large as the via angle **513** or **906**) of angles formed by a generatrix of the first hole and a generatrix of the second hole is smaller in the thin region of the base member **904** than in the thick region of the base member **904**.
(186) The base member **904** includes the second vias **905** for electrically connecting one surface of the primary surface (the base surface) **901** of the base member **904**, and the other surface of the primary surface (the base surface) **901** on the opposite side of the one surface.
(187) In each of the second vias **905**, a shape at a lower end of a first hole having a polyangular frustum shape and a shape at an upper end of a second hole having a polyangular frustum shape are the same as each other.
(188) Each of the second vias **905** has a structure in which a pattern is formed from one surface to the other surface of the primary surface (the base surface) **901** on the internal wall of the hole **908** having a shape in which the first hole and the second hole are stacked in such a manner that the lower end of the first hole and the upper end of the second hole are in contact with each other.
(189) The first hole and the second hole have the polyangular frustum shapes becoming wider as getting closer to one surface and the other surface of the primary surface (the base surface) **901** from a portion at which the lower end of the first hole and the upper end of the second hole are in contact with each other.
(190) The smaller one (which is twice as large as the via angle **513** or **906**) of angles formed by a side surface of the first hole and a side surface of the second hole that is in contact with the side surface of the first hole is smaller in the thin region of the base member **904** than in the thick region of the base member **904**.
(191) A third exemplary embodiment of the present disclosure will be described next with reference to FIGS. **9**A to **9**D.
(192) In the second exemplary embodiment, it has been described that, in a case where the via forming surface is thick, changing the via angle **906** to an obtuse angle is effective from the aspect of preventing the pattern break.
(193) In the present exemplary embodiment, an example where a recessed shape is provided near a hole **1002** of each of vias **1000** formed in a base member **1001** of the shake detection unit **320** will be described.
(194) FIG. **9**A is a cross-sectional view of each of the vias **1000** formed in the base member **1001**. FIGS. **9**B and **9**C are cross-sectional perspective views of each of the vias **1000** formed in the base member **1001**.
(195) FIG. **9**D is a schematic diagram of pattern wiring formed around each of the vias **1000**.
(196) The vias **1000** are similar to the vias **520** according to the first exemplary embodiment in that the hole **1002** is formed into a shape in which two frustums are stacked.
(197) However, the vias **1000** are different from the vias **520** in that a recessed shape is provided around each of the vias **1000** in the base member **1001**.
(198) Because the base member **1001** is shaped in this way, the base member **1001** at the via arrangement portion is locally thin, and stress applied to a pattern at a via vertex **1003** can be alleviated.

(199) The recessed shape may be a recessed shape **1004** illustrated in FIG. **9**B or **9**C, but is not limited thereto. Another effect of the provision of the recessed shape **1004** is reduction of a hole diameter of the hole **1002** of each of the vias **1000** on the surface of the base member **1001**.
(200) This effect ensures a wider wiring area on the surface of the base member **1001**.
(201) This is because another signal line **1006** can also be arranged near a signal line **1005** electrically connecting with the vias **1000**, as illustrated in FIG. **9**D.
(202) The present exemplary embodiment will be schematically described next.
(203) The base member **1001** is an injection-molded component in which pattern wiring is directly formed. In other words, the base member **1001** is an MID.
(204) The base member **1001** includes the vias **1000** for electrically connecting one surface of the primary surface (the base surface) **503**Y of the base member **1001**, and the other surface of the primary surface (the base surface) **503**Y on the opposite side of the one surface.
(205) In each of the vias **1000**, a shape at a lower end of a first hole having a frustum shape and a shape at an upper end of a second hole having a frustum shape are the same as each other.
(206) Each of the vias **1000** has a structure in which a pattern is formed from one surface to the other surface of the primary surface (the base surface) **503**Y on the internal wall of the hole **1002** having a shape in which the first hole and the second hole are stacked in such a manner that the lower end of the first hole and the upper end of the second hole are in contact with each other.
(207) The first hole and the second hole have the frustum shapes becoming wider as getting closer to one surface and the other surface of the primary surface (the base surface) **503**Y from a portion at which the lower end of the first hole and the upper end of the second hole are in contact with each other.
(208) A thickness of the base member **1001** at the recessed shape **1004** around a portion where each of the vias **1000** is arranged is thinner than a thickness of the base member **1001** in a portion where the vias **1000** are not arranged.
(209) A fourth exemplary embodiment of the present disclosure will be described next with reference to FIGS. **10**A and **10**B.
(210) In the present exemplary embodiment, an example where a hole **1100** of each of vias **1107** formed in a base member **1101** of the shake detection unit **320** has a shape in which frustums are not stacked symmetrically about a center in a thickness direction of the base member **1101** will be described.
(211) FIG. **10**A is a cross-sectional perspective view of the base member **1101** near the vias **1107** directly formed in the base member **1101**. FIG. **10**B is a cross-sectional view of the base member **1101** near the vias **1107**.
(212) In the hole **1100**, assume that a distance from a via vertex **1106** to one surface of the base member **1101** is L1, and a distance from the via vertex **1106** to the other surface of the base member **1101** is L2.
(213) In this case, two frustum shapes are formed at a via angle **1103** that satisfies the relation of $L2>L1$.
(214) In the present exemplary embodiment, with this configuration, a position of the via vertex **1106** where stress concentration occurs is displaced from the center in the thickness direction of the base member **1101** at which the stress is largest, so that the stress to be applied to a pattern at the via vertex **1106** can be alleviated.
(215) While a via diameter **1105** on one surface of the base member **1101** is larger, a via diameter **1104** on the other surface of the base member **1101** can be made smaller.
(216) Thus, arranging the side having the smaller via diameter **1104** on a side having a higher pattern wiring density, pattern wiring efficiency can be enhanced.
(217) The present exemplary embodiment will be schematically described next.
(218) The base member **1101** is an injection-molded component in which pattern wiring is directly formed.

(219) The base member **1101** includes the vias **1107** for electrically connecting one surface of the primary surface (the base surface) **503**Y of the base member **1101**, and the other surface of the primary surface (the base surface) **503**Y on the opposite side of the one surface.
(220) In each of the vias **1107**, a shape at a lower end of a first hole having a frustum shape and a shape at an upper end of a second hole having a frustum shape are the same as each other.
(221) Each of the vias **1107** has a structure in which a pattern is formed from one surface to the other surface of the primary surface (the base surface) **503**Y on the internal wall of the hole **1100** having a shape in which the first hole and the second hole are stacked in such a manner that the lower end of the first hole and the upper end of the second hole are in contact with each other.
(222) The first hole and the second hole have the frustum shapes becoming wider as getting closer to one surface and the other surface of the primary surface (the base surface) **503**Y from a portion at which the lower end of the first hole and the upper end of the second hole are in contact with each other.
(223) The width of the first hole in the direction vertical to the primary surface (the base surface) **503**Y, i.e., the distance L1 and the width of the second hole in the direction vertical to the primary surface (the base surface) **503**Y, i.e., the distance L2 are different from each other.
(224) When the base member **1101** is viewed from a direction where the primary surface (the base surface) **503**Y extends, a thickness of the base member **1101** at the recessed shape **1004** around a portion where each of the vias **1107** is arranged is thinner than a thickness of the base member **1101** in a portion where the vias **1107** are not arranged.
(225) A fifth exemplary embodiment of the present disclosure will be described next with reference to FIG. **11**.
(226) In FIG. **11**, illustration of the gyro sensor **501**P and the passive elements **502**R such as a resistor and a capacitor is omitted for ease of understanding.
(227) <Enlarged View of Gyro Sensor Mounting Lands on First Side Wall>
(228) FIG. **11** is an enlarged view of the lands **522** for mounting the gyro sensor **501**P that are formed on the first side wall **503**P.
(229) The gyro sensor **501**P is mounted onto the first side wall **503**P by reflow mounting using solder.
(230) In FIG. **11**, an arrow G indicates a gravity direction (a direction G) in melting a solder paste in the reflow mounting of the gyro sensor **501**P on the first side wall **503**P.
(231) In FIG. **11**, an arrow H indicates a direction (a direction H) that is orthogonal to the gravity direction and parallel to the first side wall **503**P.
(232) In the present exemplary embodiment, the gyro sensor **501**P is mounted onto the first side wall **503**P in a state where gravity acts in the direction indicated by the arrow G.
(233) Thus, when the solder melts, the mounting position may shift from a desired mounting position in the direction indicated by the arrow G, under the influence of the gravity in the direction indicated by the arrow G.
(234) In view of the foregoing, in land arrays A and B each including the five lands **522**, the amount of solder to be applied to the land array B on the lower side in FIG. **11** is made smaller than the amount of solder to be applied to the land array A on the upper side.
(235) As a result, surface tension of the solder applied to the land array A is larger than surface tension of the solder applied to the land array B.
(236) With this configuration, it is possible to prevent mounting misalignment caused by the gravity in the reflow mounting, by changing the direction of the force acting on the gyro sensor **501**P, to the reverse direction of the direction indicated by the arrow G, using the surface tension of the solder, and cancelling out the gravity acting on the gyro sensor **501**P in the direction indicated by the arrow G.
(237) In this case, a size of each of the lands **522** included in the land array B on the lower side in FIG. **11** is made smaller than a size of each of the lands **522** included in the land array A.

(238) Thus, the size of each of the lands **522** is suitable for the amount of solder to be applied, so that protrusion of the solder applied to the land array A, where the amount of solder applied is large, from the lands **522** can be prevented.
(239) The present exemplary embodiment will be schematically described next.
(240) The base member **503** is a molded component in which the pattern wiring **500** is directly formed.
(241) On different surfaces (the primary surface **503**Y and the first side wall **503**P) of the base member **503** that are not parallel to each other, land groups for solder mounting of the respective electronic components (the gyro sensors **501**Y and **501**P) are formed.
(242) In at least one of the land groups, the first land array (the land array A) in which the plurality of lands **522** is arrayed in the first direction (the direction H), and the second land array (the land array B) in which the plurality of lands **522** is arrayed in the first direction (the direction H) are arranged in the second direction (the direction G) different from the first direction.
(243) When viewed from a normal direction of a base surface of the base member **503**, the first land array (the land array A) is arranged above the second land array (the land array B) in the second direction.
(244) An area of each of the lands **522** included in the first land array (the land array A) is larger than an area of each of the lands **522** included in the second land array (the land array B).
(245) When viewed from the normal direction of the base surface of the base member **503**, the first land array (the land array A) is arranged above the second land array (the land array B) in the second direction.
(246) The amount of solder applied to the lands **522** included in the first land array (the land array A) is larger than the amount of solder applied to the lands **522** included in the second land array (the land array B).
(247) In FIG. **11**, the amount of solder applied to a region where the area of each of the lands **522** included in the first land array (the land array A) is large is larger than the amount of solder applied to a region where the area of each of the lands **522** included in the second land array (the land array B) is small.
(248) As a modified example of FIG. **11**, the present exemplary embodiment can also be applied to a configuration in which an area of each of lands included in an upper third land array and an area of each of lands included in a lower fourth land array are the same as each other.
(249) In this case, the amount of solder applied to the lands included in the upper third land array is larger than the amount of solder applied to the lands included in the lower fourth land array.
(250) The action and effects described with reference to FIG. **11** can be obtained.
(251) The second direction (the direction G) is orthogonal to the first direction.
(252) On different surfaces (the primary surface **503**Y and the first side wall **503**P) of the base member **503** that are not parallel to each other, land groups for reflow mounting of the respective electronic components (the gyro sensors **501**Y and **501**P) are formed.
(253) The second direction (the direction G) is the gravity direction in melting the solder in the reflow mounting.
(254) A sixth exemplary embodiment of the present disclosure will be described next with reference to FIG. **12**.
(255) <Enlarged View of Gyro Sensor Mounting Lands on First Side Wall>
(256) In FIG. **12**, illustration of the gyro sensor **501**P and the passive elements **502**R such as a resistor and a capacitor is omitted for ease of understanding.
(257) FIG. **12** is an enlarged view of lands **1302** for mounting the gyro sensor **501**P that are formed on the first side wall **503**P.
(258) The gyro sensor **501**P is mounted onto the first side wall **503**P by reflow mounting using solder.
(259) In FIG. **12**, an arrow G indicates a gravity direction (a direction G) in melting a solder paste

in the reflow mounting of the gyro sensor **501**P on the first side wall **503**P.
(260) In addition, an arrow H indicates a direction (a direction H) that is orthogonal to the gravity direction and parallel to the first side wall **503**P.
(261) In the present exemplary embodiment, the gyro sensor **501**P is mounted onto the first side wall **503**P in a state where gravity acts in the direction indicated by the arrow G.
(262) Thus, when the solder melts, the mounting position may shift from a desired mounting position in the direction indicated by the arrow G, under the influence of the gravity in the direction indicated by the arrow G.
(263) In the present exemplary embodiment, a width of each of the ten lands **1302** in total is varied in the direction indicated by the arrow G.
(264) More specifically, each of the lands **1302** is formed into a shape in which the width becomes smaller toward the gravity direction.
(265) As the width becomes larger, the force acting on the gyro sensor **501**P increases with the surface tension of the solder.
(266) Thus, by forming the lands **1302** into the above-described shape, the direction of the force acting on the gyro sensor **501**P is changed to the reverse direction of the direction indicated by the arrow G, using the surface tension of the solder.
(267) Cancelling out the gravity acting on the gyro sensor **501**P in the direction indicated by the arrow G can prevent mounting misalignment caused by the gravity in the reflow mounting.
(268) The present exemplary embodiment will be schematically described next.
(269) The base member **503** is a molded component in which the pattern wiring **500** is directly formed.
(270) On different surfaces (the primary surface **503**Y and the first side wall **503**P) of the base member **503** that are not parallel to each other, land groups for solder mounting of the respective electronic components (the gyro sensors **501**Y and **501**P) are formed.
(271) When viewed from a normal direction of a base surface of the base member **503**, at least one of the land groups includes a land array in which the plurality of lands **1302** is arranged in the predetermined direction (the direction H).
(272) When each of the lands **1302** is viewed from the normal direction of the base surface of the base member **503**, in the direction G orthogonal to the predetermined direction (the direction H), the lower width of each of the lands **1302** in the predetermined direction (the direction H) is smaller than the upper width of each of the lands **1302** in the predetermined direction (the direction H).
(273) On different surfaces (the primary surface **503**Y and the first side wall **503**P) of the base member **503** that are not parallel to each other, land groups for reflow mounting of the respective electronic components (the gyro sensors **501**Y and **501**P) are formed.
(274) The direction G orthogonal to the predetermined direction (the direction H) is the gravity direction in melting the solder in the reflow mounting.
(275) A seventh exemplary embodiment of the present disclosure will be described next with reference to FIG. **13**.
(276) In FIG. **13**, illustration of the gyro sensor **501**P and the passive elements **502**R such as a resistor and a capacitor is omitted for ease of understanding.
(277) <Enlarged View of Gyro Sensor Mounting Lands on First Side Wall>
(278) FIG. **13** is an enlarged view of lands **1402** for mounting the gyro sensor **501**P that are formed on the first side wall **503**P.
(279) The gyro sensor **501**P is mounted onto the first side wall **503**P by reflow mounting using solder.
(280) In FIG. **13**, a shape of the gyro sensor **501**P is schematically indicated by broken lines.
(281) Mounting terminals **1403** are connection terminals for mounting the gyro sensor **501**P onto the lands **1402**.

(282) In FIG. **13**, positions and shapes of the mounting terminals **1403** are schematically indicated by broken lines.
(283) In FIG. **13**, an arrow G indicates a gravity direction (a direction G) in melting a solder paste in the reflow mounting of the gyro sensor **501**P on the first side wall **503**P.
(284) In addition, an arrow H indicates a direction (a direction H) that is orthogonal to the gravity direction and parallel to the first side wall **503**P.
(285) In the present exemplary embodiment, the gyro sensor **501**P is mounted onto the first side wall **503**P in a state where gravity acts in the direction indicated by the arrow G.
(286) Thus, when the solder melts, the mounting position may shift from a desired mounting position in the direction indicated by the arrow G, under the influence of the gravity acting in the direction indicated by the arrow G.
(287) In the present exemplary embodiment, a width of each of the ten lands **1402** in total is varied in the direction indicated by the arrow G.
(288) More specifically, in the direction indicated by the arrow G, a width of each of the lands **1402** at a position corresponding to an end of the corresponding mounting terminal **1403** on an outer side of the gyro sensor **501**P in the direction indicated by the arrow G is made widest.
(289) Each of the lands **1402** is formed into a shape becoming narrower as getting away from the position corresponding to the end of the mounting terminal **1403** in the direction indicated by the arrow G.
(290) The position where the width of each of the lands **1402** is largest is a position where the force acting on the gyro sensor **501**P is largest with the surface tension of the solder.
(291) Thus, in the reflow mounting, the end of each of the mounting terminals **1403** of the gyro sensor **501**P located on the outer side of the gyro sensor **501**P is kept at the position at which the width of each of the lands **1402** is largest, so that mounting misalignment caused by the gravity can be prevented.
(292) The present exemplary embodiment will be schematically described next.
(293) The base member **503** is a molded component in which the pattern wiring **500** is directly formed.
(294) On different surfaces (the primary surface **503**Y and the first side wall **503**P) of the base member **503** that are not parallel to each other, land groups for solder mounting of the respective electronic components (the gyro sensors **501**Y and **501**P) are formed.
(295) When viewed from a normal direction of a base surface of the base member **503**, at least one of the land groups includes a land array A or B in which the plurality of lands **1402** is arrayed in the predetermined direction (the direction H).
(296) Each of the lands **1402** has a structure in which, in the predetermined direction (the direction H), the portion corresponding to the end of the corresponding mounting terminal **1403** of the electronic component (the gyro sensor **501**P) that is on the outer side of the electronic component (the gyro sensor **501**P) has the largest width.
(297) In the direction G orthogonal to the predetermined direction (the direction H), each of the lands **1402** has a smaller width toward an upward direction from the portion corresponding to the end of the corresponding mounting terminal **1403** of the electronic component (the gyro sensor **501**P) that is on the outer side of the electronic component (the gyro sensor **501**P).
(298) When viewed from the normal direction of the base surface of the base member **503**, in the direction G orthogonal to the predetermined direction (the direction H), each of the lands **1402** has a smaller width toward a downward direction from the portion corresponding to the end of the corresponding mounting terminal **1403** of the electronic component (the gyro sensor **501**P) that is on the outer side of the electronic component (the gyro sensor **501**P).
(299) On different surfaces (the primary surface **503**Y and the first side wall **503**P) of the base member **503** that are not parallel to each other, land groups for reflow mounting of the respective electronic components (the gyro sensors **501**Y and **501**P) are formed.

(300) The direction G orthogonal to the predetermined direction (the direction H) is the gravity direction in melting the solder in the reflow mounting.
(301) An eighth exemplary embodiment of the present disclosure will be described next with reference to FIGS. **14**A and **14**B.
(302) <Base Member with Pattern Wiring and Mounting Component)
(303) FIGS. **14**A and **14**B illustrate a base member **1500** having pattern wiring **1502** and a mounting component **1501** that are similar to those of the base member **503** described in the first exemplary embodiment.
(304) On the base member **1500**, the pattern wiring **1502** is directly formed, and the mounting component **1501** is directly mounted.
(305) In FIGS. **14**A and **14**B, an arrow G indicates a gravity direction (a direction G) in melting a solder paste in reflow mounting of the gyro sensor **501**P on the first side wall **503**P.
(306) In addition, an arrow H indicates a direction (a direction H) that is orthogonal to the gravity direction and parallel to the first side wall **503**P.
(307) On the base member **1500** on the direction G side of the mounting component **1501**, standing walls **1503** having a given height are arranged in the direction H so as to be in contact with the mounting component **1501**.
(308) Thus, when the solder melts in a mounting process, it is possible to prevent the mounting component **1501** from being displaced or dropping due to the gravity.
(309) In the present exemplary embodiment, the example where the standing walls **1503** are provided in a divided manner so as to avoid contact with the pattern wiring **1502** directly wired to the base member **1500** has been described.
(310) Alternatively, the standing walls **1503** may be continuous without being provided in a divided manner, and the pattern wiring **1502** may be formed so as to cross the standing walls **1503**.
(311) While the exemplary embodiments of the present disclosure have been described above, the exemplary embodiments are not limited to the configurations described in the exemplary embodiments.
(312) Materials, shapes, dimensions, configurations, numbers, and arrangement positions can be appropriately changed without departing from the gist of the present disclosure.
(313) The exemplary embodiments of the present disclosure have been described using a camera and a smartphone as examples of an electronic device.
(314) The exemplary embodiments can also be applied to various electronic devices such as a personal computer, a tablet terminal, a game machine, a drone, an automobile, and peripheral devices thereof.
(315) According to the exemplary embodiments of the present disclosure, it is possible to provide a base member formed using the MID technique, and having a shape suitable for being accommodated in an electronic device.
(316) While the present disclosure has been described with reference to exemplary embodiments, it is to be understood that the disclosure is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.
(317) This application claims the benefit of Japanese Patent Application No. 2021-061120, filed Mar. 31, 2021, which is hereby incorporated by reference herein in its entirety.

Claims

1. A base member being a molded component in which pattern wiring is directly formed, the base member comprising: a base surface including a first surface and a second surface on an opposite side of the first surface, wherein the base surface includes a thin region and a thick region, and wherein a distance between the first surface and the second surface in the thin region is less than a

distance between the first surface and the second surface in the thick region; and a first via in the thin region and a second via in the thick region for electrically connecting the first surface and the second surface, wherein, in the first via, a shape at a lower end of a first hole having a conical frustum shape and a shape at an upper end of a second hole having a conical frustum shape are identical to each other, wherein when viewed from a direction in which the base surface extends, the first via has a structure in which a pattern is formed from the first surface to the second surface of the base surface on an internal wall of a fifth hole having a shape in which the first hole and the second hole are stacked in such a manner that the lower end of the first hole and the upper end of the second hole are in contact with each other, wherein, in the second via, a shape at a lower end of a third hole having a conical frustum shape and a shape at an upper end of a fourth hole having a conical frustum shape are identical to each other, wherein when viewed from a direction in which the base surface extends, the second via has a structure in which a pattern is formed from the first surface to the second surface of the base surface on an internal wall of a sixth hole having a shape in which the third hole and the fourth hole are stacked in such a manner that the lower end of the third hole and the upper end of the fourth hole are in contact with each other, wherein, when viewed from the direction in which the base surface extends, conical frustum shape of the first hole becomes wider closer to the first surface relative to a portion at which the lower end of the first hole and the upper end of the second hole are in contact with each other, wherein, when viewed from the direction in which the base surface extends, the conical frustum shape of the second hole becomes wider closer to the second surface relative to the portion at which the lower end of the first hole and the upper end of the second hole are in contact with each other, wherein, when viewed from the direction in which the base surface extends, the conical frustum shape of the third hole becomes wider closer to the first surface relative to a portion at which the lower end of the third hole and the upper end of the fourth hole are in contact with each other, wherein, when viewed from the direction in which the base surface extends, the conical frustum shape of the fourth hole becomes wider closer to the second surface relative to the portion at which the lower end of the third hole and the upper end of the fourth hole are in contact with each other, and wherein when the first hole, the second hole, the third hole, and the fourth hole are viewed from the direction in which the base surface extends, an angle formed by a generatrix of the first hole and a generatrix of the second hole, which are in the thin region of the base member, is smaller than an angle formed by a generatrix of the third hole and a generatrix of the fourth hole, which are in the thick region of the base member.

2. The base member according to claim 1, wherein, when the base member is viewed from the direction in which the base surface extends, a thickness of the thin region the base member around a portion where the first via is arranged is thinner than a thickness of the thin region the base member at a portion where the via is not arranged.

3. A base member according to claim 1, wherein a width of the first hole in a direction vertical to the base surface and a width of the second hole in the direction vertical to the base surface are different from each other.